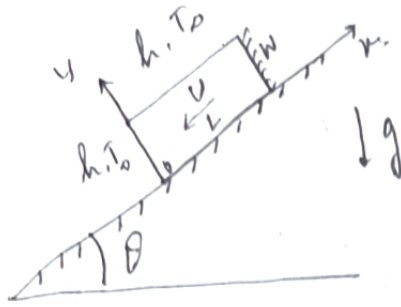


81)



$$\nabla(k \nabla T) + \dot{q} = \rho c \frac{\partial T}{\partial t}$$

no heat generation ; $\dot{q} = 0$

Steady state ; $\frac{\partial T}{\partial t} = 0$

constant thermal conductivity ; $k = f(x, y, t)$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

@ $x=0$; $-k \frac{\partial T(x, y)}{\partial y} = h(T(x, y) - T_0)$
 $x=L$; $-k \frac{\partial T}{\partial x}$

(x, W) ; $-k \frac{\partial T(x, y)}{\partial y} = h(T(x, y) - T_a)$

$(x, 0)$; $-k \frac{\partial T(x, 0)}{\partial y} = q_0''$

$(0, y)$; $-k \frac{\partial T(0, y)}{\partial x} = h(T(0, y) - T_0)$

(L, y) ; $k \frac{\partial T(L, y)}{\partial x} = 0$

8/3

Conduction Problems:

Assignment:

Q2).



$$\nabla(k \nabla T) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\nabla(k \nabla T) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\nabla^2 T + \dot{q}/k = \frac{\rho}{\alpha} \frac{\partial T}{\partial t}$$

$$\dot{q} = \dot{q}_0 e^{-\rho t}$$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\dot{q}_0}{k} e^{-\rho t} = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

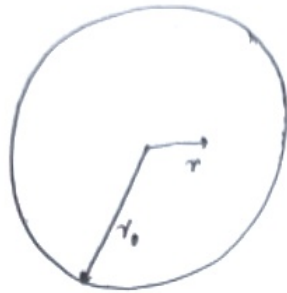
@ $t=0$, $T=T_0$

@ $x=-L$; $T=T_L(t)$

@ $x=+L$; $T=T_0$

Q3)

T_0



for spherical system,

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = \frac{1}{r^2 \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dT}{d\theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{d^2 T}{d\phi^2}$$

$$\rightarrow \dot{q}/h = \frac{1}{\alpha} \frac{dT}{dr}$$

1-D flow.

$T = T(r)$ only.

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dT}{dr} \right) = \dot{q}/h = \frac{1}{\alpha} \frac{dT}{dr}$$

@ $t = 0$; $T = T_0$ $\dot{q} = 0$

@ $r = 0$; $T = \text{finite}$

$r = r_0$; $-k \frac{dT}{dr} = h(T - T_0)$

$$\theta = \frac{T - T_0}{T_0 - T_0}$$

$$\lambda = \frac{r}{r_0}$$

$$\frac{\delta \theta}{\delta \lambda} = \frac{\delta T}{\delta r} \cdot \frac{r}{r_0} \cdot \frac{\delta r}{\delta \lambda}$$

$$= \frac{\delta T}{\delta r} \cdot r_0 \cdot \frac{1}{(T_0 - T_\infty)}$$

$$\frac{\delta T}{\delta r} = \frac{\delta \theta}{\delta r} \cdot \frac{\delta r}{\delta \theta}$$

$$= (T_0 - T_\infty) \frac{\delta \theta}{\delta r}$$

$$\frac{\delta T}{\delta r} = \frac{(T_0 - T_\infty)}{r_0} \cdot \frac{\delta \theta}{\delta \lambda}$$

$$r_0 \frac{\delta T}{\delta r} = r_0 \cdot \frac{(T_0 - T_\infty)}{r_0} \cdot \frac{\delta \theta}{\delta \lambda}$$

$$\frac{\delta}{\delta r} \left(r_0 \frac{\delta T}{\delta r} \right) = \frac{\delta}{\delta \lambda} \left(r_0 \frac{(T_0 - T_\infty)}{r_0} \cdot \frac{\delta \theta}{\delta \lambda} \right) \cdot \frac{\delta \lambda}{\delta r} \cdot \frac{1}{r_0}$$

$$= \frac{\delta}{\delta \lambda} \left(\lambda^2 \cdot \frac{(T_0 - T_\infty)}{r_0} \cdot \frac{\delta \theta}{\delta \lambda} \right)$$

$$= \frac{\delta}{\delta \lambda} \left(\lambda^2 (T_0 - T_\infty) \frac{\delta \theta}{\delta \lambda} \right)$$

$$\frac{1}{r_0} \frac{\delta}{\delta r} \left(r_0 \frac{\delta T}{\delta r} \right) = \frac{1}{\lambda^2 r_0} \cdot \frac{\delta}{\delta \lambda} \left[\lambda^2 (T_0 - T_\infty) \frac{\delta \theta}{\delta \lambda} \right]$$

$$= \frac{(T_0 - T_\infty)}{r_0} \cdot \frac{1}{\lambda^2} \frac{\delta}{\delta \lambda} \left[\lambda^2 \frac{\delta \theta}{\delta \lambda} \right]$$

$$\frac{(T_0 - T_\infty)}{r_0} \cdot \frac{1}{\lambda^2} \frac{\delta}{\delta \lambda} \left(\lambda^2 \frac{\delta \theta}{\delta \lambda} \right) = \frac{1}{\alpha} \cdot \frac{(T_0 - T_\infty)}{r_0} \frac{\delta \theta}{\delta r}$$

$$\frac{1}{\lambda^2} \frac{\delta}{\delta \lambda} \left(\lambda^2 \frac{\delta \theta}{\delta \lambda} \right) = \frac{r_0}{\alpha} \cdot \frac{\delta \theta}{\delta r}$$

$$\frac{1}{\lambda^2} \frac{\delta}{\delta \lambda} \left(\lambda^2 \frac{\delta \theta}{\delta \lambda} \right) = \frac{\delta \theta}{\delta r}$$

$$\left[\lambda^2 \frac{\delta^2 \theta}{\delta \lambda^2} + 2 \lambda \frac{\delta \theta}{\delta \lambda} \right]$$

$$\text{C } \lambda^2 = 0; \quad \text{D } = 1$$

$$\text{C } \lambda^2 = 0; \quad \text{D } = h \cdot h$$

$$\text{C } \lambda^2 = 1; \quad -h \cdot \left(\frac{T_0 - T_\infty}{r_0} \right) \cdot \frac{\delta \theta}{\delta \lambda} = h (T_0 - T_\infty)$$

$$\frac{\delta \theta}{\delta \lambda} + h r_0 = 0$$

$$\frac{\partial \theta}{\partial T} = \frac{\partial \theta}{\partial T} \cdot \frac{\partial T}{\partial \lambda} \cdot \frac{\partial \lambda}{\partial T}$$

$$= \frac{\partial \theta}{\partial T} \cdot \frac{1}{(T_0 - T_0)}$$

$$\frac{\partial T}{\partial \lambda} = \frac{(T_0 - T_0)}{T_0} \cdot \frac{\partial \theta}{\partial \lambda}$$

$$= \frac{\partial \theta}{\partial \eta} \cdot \frac{\partial \eta}{\partial \lambda}$$

$$\frac{\partial \theta}{\partial T} = \frac{\partial \theta}{\partial \lambda} \cdot \frac{\partial \lambda}{\partial T}$$

$$= \frac{\partial \theta}{\partial \lambda} \cdot \left(\frac{\partial \lambda}{\partial T} \cdot \frac{\partial T}{\partial \eta} \right) \cdot \frac{\partial \eta}{\partial T}$$

$$= \frac{\partial \theta}{\partial \lambda} \cdot \left(\frac{\partial \lambda}{\partial T} \cdot \frac{\partial T}{\partial \eta} \cdot \frac{\partial \eta}{\partial T} \right)$$

$$= \frac{\partial \theta}{\partial \lambda} \cdot \left(\frac{\partial \lambda}{\partial T} \cdot \frac{\partial T}{\partial \eta} \cdot \frac{\partial \eta}{\partial T} \right)$$

$$\frac{\partial T}{\partial \lambda} = \frac{\partial \theta}{\partial \lambda} \cdot \frac{\partial \lambda}{\partial T}$$

$$= (T_0 - T_0) \cdot \frac{\partial \theta}{\partial \lambda}$$